

BLOCK CIPHER



Symmetric Key Encryption

ALICE



BOB



- Block ciphers and stream ciphers are two **primitives** of symmetric key cryptosystems

Encryption

- Let X be plaintext
- Let F be the round function for all the round
- The encryption operation is carried out as follows:

$$\begin{aligned}X_0 &\leftarrow X \\X_1 &\leftarrow F(X_0, K_1) \\X_2 &\leftarrow F(X_1, K_2) \\&\vdots \\X_r &\leftarrow F(X_{r-1}, K_r)\end{aligned}$$

- Ciphertext $Y = X_r$



Decryption

- F is injective function if its second argument is fixed.
- There exists a function F^{-1} such that $F^{-1}(F(X, Y), Y) = X$. Then the decryption operation is carried out as follows:

$$\begin{aligned} X_r &\leftarrow Y \\ X_{r-1} &\leftarrow F^{-1}(X_r, K_r) \\ &\vdots \\ X_1 &\leftarrow F^{-1}(X_2, K_2) \\ X_0 &\leftarrow F^{-1}(X_1, K_1) \\ X &\leftarrow X_0 \end{aligned}$$



Substitution-Permutation Networks

We begin by defining a *substitution-permutation network*, or *SPN*. (An SPN is a special type of iterated cipher with a couple of small changes that we will indicate.) Suppose that ℓ and m are positive integers. A plaintext and ciphertext will both be binary vectors of length ℓm (i.e., ℓm is the *block length* of the cipher). An SPN is built from two components, which are denoted π_S and π_P .

$$\pi_S : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$$

is a permutation of the 2^ℓ bitstrings of length ℓ , and

$$\pi_P : \{1, \dots, \ell m\} \rightarrow \{1, \dots, \ell m\}$$

is also a permutation, of the integers $1, \dots, \ell m$. The permutation π_S is called an *S-box* (the letter “S” denotes “substitution”). It is used to replace ℓ bits with a different set of ℓ bits. π_P , on the other hand, is used to permute ℓm bits by changing their order.

- **Key Scheduling Algorithm:** $K = (k_1, \dots, k_{32})$. For $1 \leq i \leq 5$, define K_i to consist of 16 consecutive bits of K , beginning with k_{4i-3} .
- Example: Suppose the key is

$k = 0011 \ 1010 \ 1001 \ 0100 \ 1101 \ 0110 \ 0011 \ 1111$

- The the round keys are as follows:

$$K_1 = 0011 \quad 1010 \quad 1001 \quad 0100$$

$$K_2 = 1010 \quad 1001 \quad 0100 \quad 1101$$

$$K_3 = 1001 \quad 0100 \quad 1101 \quad 0110$$

$$K_4 = 0100 \quad 1101 \quad 0110 \quad 0011$$

$$K_5 = 1101 \quad 0110 \quad 0011 \quad 1111$$



- Let $l = m = r = 4$ and π_S be defined as follows, where the input (i.e., z) and the output (i.e., $\pi_S(z)$) are written in hexadecimal notation,
 $(0 = (0, 0, 0, 0), 1 = (0, 0, 0, 1), \dots, 9 = (1, 0, 0, 1), A = (1, 0, 1, 0), \dots, F = (1, 1, 1, 1))$:

z	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$\pi_S(z)$	E	4	D	1	2	F	B	8	3	A	6	C	5	9	0	7

- Let π_P be defined as follows:

z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$\pi_P(z)$	1	5	9	13	2	6	10	14	3	7	11	15	4	8	12	16



- Suppose that the plaintext is

$$X = 0010 \ 0110 \ 1011 \ 0111$$

- Then the encryption of X proceeds as follows:

$$w^0 = 0010 \quad 0110 \quad 1011 \quad 0111$$

$$K_1 = 0011 \quad 1010 \quad 1001 \quad 0100$$

$$u^1 = 0001 \quad 1100 \quad 0010 \quad 0011$$

$$v^1 = 0100 \quad 0101 \quad 1101 \quad 0001$$

$$w^1 = 0010 \quad 1110 \quad 0000 \quad 0111$$

$$K_2 = 1010 \quad 1001 \quad 0100 \quad 1101$$

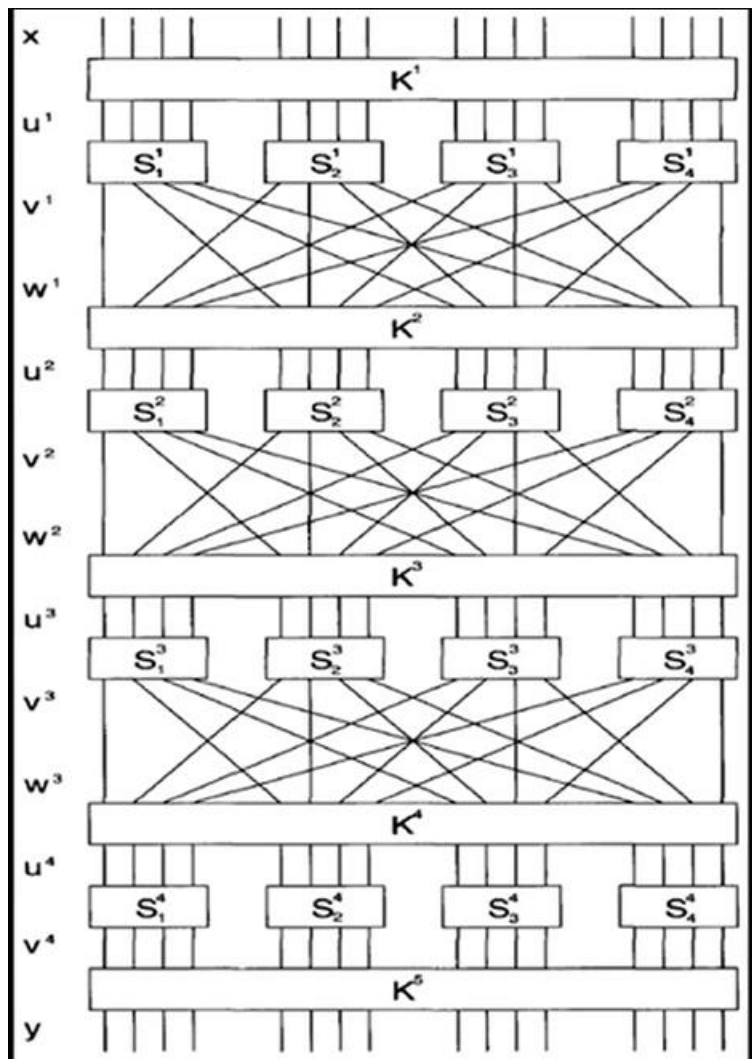
$$u^2 = 1000 \quad 0111 \quad 0100 \quad 1010$$





v^2	=	0011	1000	0010	0110
w^2	=	0100	0001	1011	1000
K_3	=	1001	0100	1101	0110
u^3	=	1101	0101	0110	1110
v^3	=	1001	1111	1011	0000
w^3	=	1110	0100	0110	1110
K_4	=	0100	1101	0110	0011
u^4	=	1010	1001	0000	1101
v^4	=	0110	1010	1110	1001
K_5	=	1101	0110	0011	1111
Y	=	1011	1100	1101	0110

- Y is the ciphertext



Data Encryption Standard (DES)

- DES is a 16-round Feistel cipher having block length 64: it encrypts a plaintext bitstring x (of length 64) using a 56-bit key, K , obtaining a ciphertext bitstring (of length 64).
- DES is the most widely used encryption scheme, adopted in 1977 by the National Bureau of Standards, now National Institute of Standards and Technology (NIST).
- Historically DES resulted from a project first initiated by IBM in the 60's which resulted in LUCIFER, a block cipher (64 bits) which used a key size of 128 bits. The NSA got involved and the final product, DES, ended up with a 56 bit key!

IP							
58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

(a) Initial permutation IP

IP^{-1}							
40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

(b) Final permutation IP^{-1}

E					
32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

Expansion permutation (E)

S-box S_1

S_1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	01	10	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

S-box S_2

S_2	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	15	01	08	14	06	11	03	04	09	07	02	13	12	00	05	10
1	03	13	04	07	15	02	08	14	12	00	01	10	06	09	11	05
2	00	14	07	11	10	04	13	01	05	08	12	06	09	03	02	15
3	13	08	10	01	03	15	04	02	11	06	07	12	00	05	14	09

S-box S_3

10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S-box S_4

S_4	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	07	13	14	03	00	06	09	10	01	02	08	05	11	12	04	15
1	13	08	11	05	06	15	00	03	04	07	02	12	01	10	14	09
2	10	06	09	00	12	11	07	13	15	01	03	14	05	02	08	04
3	03	15	00	06	10	01	13	08	09	04	05	11	12	07	02	14

S-box S_5

S_5	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	02	12	04	01	07	10	11	06	08	05	03	15	13	00	14	09
1	14	11	02	12	04	07	13	01	05	00	15	10	03	09	08	06
2	04	02	01	11	10	13	07	08	15	09	12	05	06	03	00	14
3	11	08	12	07	01	14	02	13	06	15	00	09	10	04	05	03

S-box S_6

S_6	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	12	01	10	15	09	02	06	08	00	13	03	04	14	07	05	11
1	10	15	04	02	07	12	09	05	06	01	13	14	00	11	03	08
2	09	14	15	05	02	08	12	03	07	00	04	10	01	13	11	06
3	04	03	02	12	09	05	15	10	11	14	01	07	06	00	08	13

S-box S_7

S_7	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	04	11	02	14	15	00	08	13	03	12	09	07	05	10	06	01
1	13	00	11	07	04	09	01	10	14	03	05	12	02	15	08	06
2	01	04	11	13	12	03	07	14	10	15	06	08	00	05	09	02
3	06	11	13	08	01	04	10	07	09	05	00	15	14	02	03	12

S-box S_8

S_8	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	13	02	08	04	06	15	11	01	10	09	03	14	05	00	12	07
1	01	15	13	08	10	03	07	04	12	05	06	11	00	14	09	02
2	07	11	04	01	09	12	14	02	00	06	10	13	15	03	05	08
3	02	01	14	07	04	10	08	13	15	12	09	00	03	05	06	11

P							
16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

permutation function (P)

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

Input Key

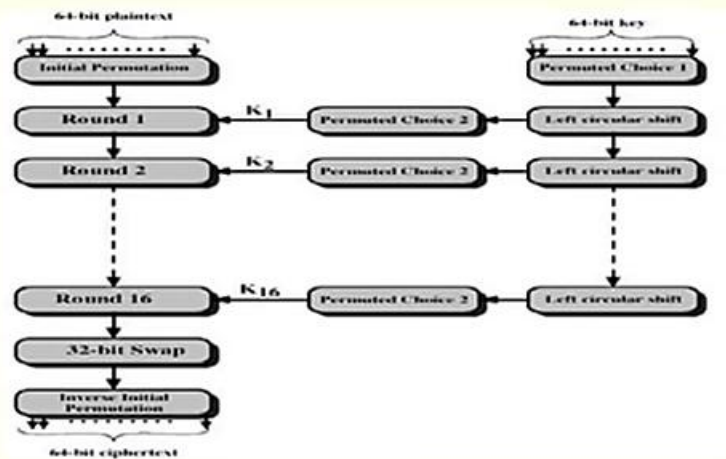
57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22

Round number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bits rotated	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

Schedule of Left Shifts

<i>PC – 2</i>							
14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

Permutation Choice Two (*PC – 2*)



Block diagram of DES algorithm.